

[JEWZ](#): Jacob Lukose, Ethan Sie, Will Zhang, Ziyad Hamed
SoftDev
P04: Makers Makin' It, Act II – The Seequel
2025-03-26
Time Spent: 2
TARGET SHIP DATE: 2025-04-07

DESIGN DOCUMENT (V1)

I. Description

This project is a website that contains a list of stocks in the S&P 500 that you can view graphically to see its fluctuation in the stock market. We show various other graphs such as comparisons between certain stocks we want. We will use machine learning to predict the future of some of the stocks if users want “financial advice”. More niche and in depth functionalities explained further in the doc.

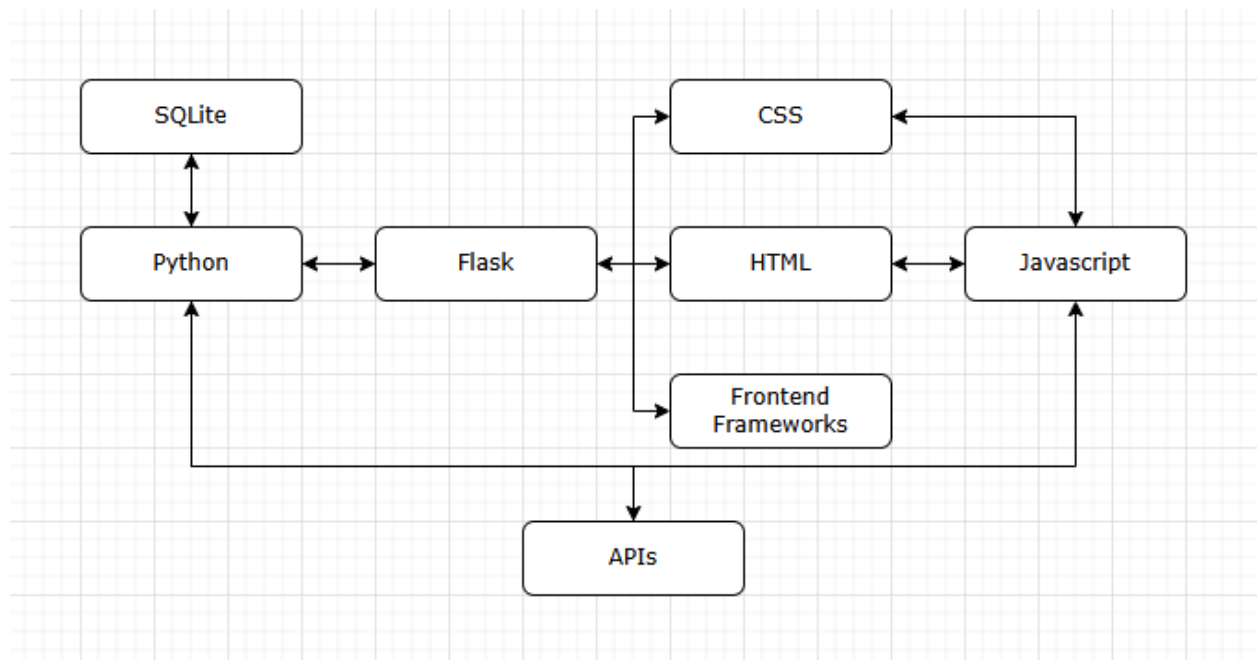
A. Main Website Functionalities

1. Intra-user Functions
 - a. Graph datasets of stocks
 - b. Apply filters to analyze the stock visuals
 - c. Predict stock growth/decline
 - d. Relevance of a stock in the news at a certain time (News API)
 - e. Follow Stocks
 - f. Light and Dark mode
 - g. Widgets
2. Inter-user Functions
 - a. Follow other Traders to see what stocks they're following
 - b. Stock battle mode
 - i. Draw your stocks to battle
 - ii. Which stock do you think will grow the most

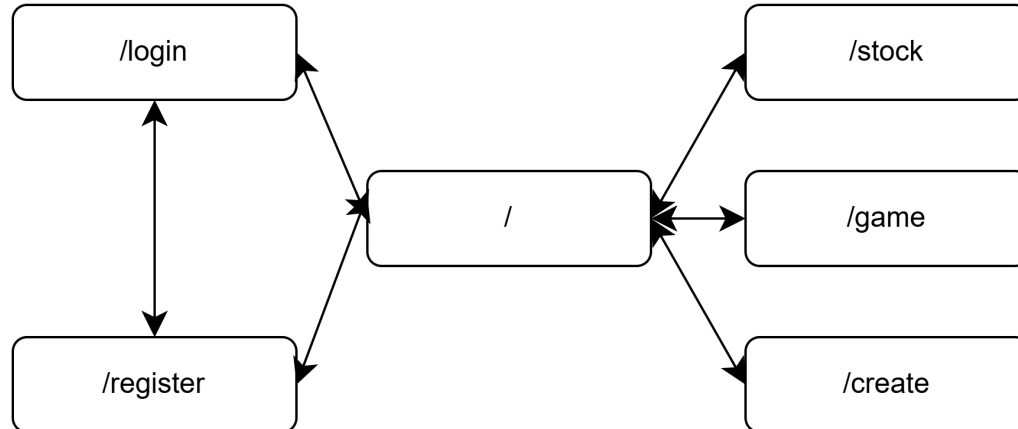
B. Program Components

- a. User Accounts:
 - i. Creation of accounts and login/logout functionality
 - ii. Sessions
- b. Routes to different pages of the website using Flask and Python

- c. SQLite3 Database: Stores data of the username and password and possibly data received from the csv file/dataset so that we do not have to read the csv file every time we reload the site.
- d. Jinja Templates:
 - i. User dashboard:
 - 1. Shows a list of companies with their current price, short_name, and maybe their logo.
 - a. If you have a watch list, it'll show the companies you have added to it.
 - 2. List of news articles that can help you decide what stocks you want to invest in.
 - 3. Shows login/sign-up
 - 4. Options to choose companies to analyze graphically or predict future earnings
 - ii. Page with graphs of selected companies and up to date news articles about those companies.
- e. JavaScript Components: Used to make graphs more interactive. Possible trace the graph and its dates to see specific prices at certain times.



C. Site Map + Descriptions



- / will contain the user dashboard explained above.
- /login and /register are self explanatory. Same or similar to previous projects. May be required for all functionalities such as tracking high scores for /game or following different stocks.
- /stock will show the graph of each stock in our dataset over the past 5 years. It will also show any recent news articles associated with that stock. Other interactive features too hopefully like following the stock or adding it to your watchlist.
- /game will be either
 - Show graphs of 2 companies over a year or so and guess which company did better the next year. Similar to the higher/lower game about google searches.
 - Or draw your own graph such that if AI is trained, they would predict the most growth in the future of the company that graph would portray.
- Draw your own graph would be part of /create. Day as a pump and dump influencer or Elon Musk where you get to control the market and artificially boost your own stock.

D. Database Organization

Database Organization:

1. users table
 - a. user_id (integer): unique identifier per user
 - b. username (string): username chosen by user
 - c. password (string): hashed password for security
 - d. last_login (string {date-time}): tracks last user interaction
 - e. created_at (string {date-time}): tracks time of creation
2. stockFollowTrack table
 - a. user_id (integer): unique identifier per user
 - b. stock_id (string): unique identifier per stock
3. customStock table
 - a. user_id (integer): unique identifier per user

- b. stock_id (string): unique identifier per stock
- 4. stockInfo table
 - a. stock_id (string): unique identifier per stock
 - b. track1 (integer): day 1 stock value
 - c. track2 (integer): day 2 stock value
 - d. track3 (integer): day 3 stock value
 - e. track4 (integer): day 4 stock value
 - f. track5 (integer): day 5 stock value
- 5. gameChallenge table
 - a. challenge_ID (integer): identifier for each challenge
 - b. challenger (string): username of the user challenging
 - c. challenged (string): username of the challenged user
 - d. accepted? (string): whether the challenge was accepted
- 6. gameHistory table
 - a. game_ID (integer): identifier for each game
 - b. winner (string): username of the winner of the game
 - c. loser (string): username of the loser of the game
 - d. time_started (string {date-time}): timestamp of the start of the game
 - e. time_completed (string {date-time}): timestamp of the end of the game

E. APIS

1. Yahoo Finance API to convert ticker symbol to full name.
2. [Stock News API](#)

F. Frontend Framework

Tailwind

Tailwind is convenient to use and easy to implement, and we seek to use Tailwind to optimize our development speed.

G. Visualization Library

Apex Charts

We plan to use Apex Charts because of its easy to implement charts and aesthetic nature that would work well with visualizing stocks.

H. Task Breakdown

Jacob Lukose -

- Project Manager
- Frontend Framework

- HTML + CSS + Jinja

Ethan Sie -

- Database Engineer
- Backend

Yinwei Zhang -

- Javascript
- Flask Middleware

Ziyad Hamed -

- API Handling